



Session 08: Slicing the Number Line — Mastering Inequalities

Phase 2 — Classical Techniques | 45 min

Part A: Marking Ranges on the Number Line

Example 1: Linear Inequalities — Only Careful When Dividing by Negatives

$$3x + 5 > 2x - 1.$$

- ① Gather x terms on the left. Move $2x$ to the left $\rightarrow 3x - 2x + 5 > -1$.
- ② Gather numbers on the right. Move 5 to the right $\rightarrow x > -6$.

$$-2x < 6.$$

- ① Divide both sides by -2 .
- ② **Dividing by a negative flips the inequality sign.** $\rightarrow x > -3$.

$$-3 < 2x + 1 \leq 5.$$

- ① Subtract 1 $\rightarrow -4 < 2x \leq 4$.
- ② Divide by 2 $\rightarrow -2 < x \leq 2$.

Draw on the number line: empty circle at -2 , filled circle at 2 , shade between them.

Example 2: Quadratic Inequalities — Tear, Split into Intervals, Check Signs

$$x^2 - 5x + 6 > 0.$$

- ① Tear first: $(x - 2)(x - 3) > 0$.
- ② Find the points where it hits 0: $x = 2, x = 3$.
- ③ These two points split the number line into three pieces.
- ④ Pick one number from each interval and check the sign.
 - $x < 2$: plug in $x = 0 \rightarrow (-2)(-3) = 6 > 0$. ✓
 - $2 < x < 3$: plug in $x = 2.5 \rightarrow (0.5)(-0.5) = -0.25 < 0$. ✗
 - $x > 3$: plug in $x = 4 \rightarrow (2)(1) = 2 > 0$. ✓

$$\rightarrow x < 2 \text{ or } x > 3.$$

Example 3: Negative Leading Coefficient — Flip to Positive First

$$-x^2 + 4x - 3 \geq 0.$$

- ① Multiply by -1 and flip the inequality $\rightarrow x^2 - 4x + 3 \leq 0$.
 - ② Tear: $(x - 1)(x - 3) \leq 0$.
 - ③ Zero points: $x = 1, x = 3$. Three intervals.
 - $x < 1$: $(-)(-) = +$. ✗
 - $1 < x < 3$: $(+)(-) = -$. ✓
 - $x > 3$: $(+)(+) = +$. ✗
 - ④ The boundaries $x = 1, x = 3$ give 0, so include them. $\rightarrow 1 \leq x \leq 3$.
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Example 4: Just Looking at the Discriminant

$$x^2 + x + 1 > 0.$$

- ① Try tearing: no two integers add to 1 and multiply to 1.
- ② Discriminant: $1^2 - 4 \times 1 \times 1 = -3 < 0$.
- ③ x^2 coefficient positive, discriminant negative $\rightarrow$ the parabola sits entirely above the x -axis.
 $\rightarrow$ **True for all x . Always true.**

$$x^2 + 2x + 1 \leq 0.$$

- ① $(x + 1)^2 \leq 0$.
- ② A square can't be less than 0. Only the point where it equals 0 works.
 $\rightarrow$ **True only at $x = -1$. Exactly one point.**

Part B: Rational, Higher-Degree, and Absolute Value Inequalities

Example 5: Rational Inequalities — Never Recklessly Multiply by the Denominator

$$\frac{x-1}{x+2} > 0.$$

- ① Find the points where the sign can flip.
Numerator = 0 $\rightarrow x = 1$. Denominator = 0 $\rightarrow x = -2$.
These two points split the number line into three. $x = -2$ is excluded (denominator 0).
- ② Pick one number from each interval and check the sign.
 - $x < -2$: $x = -3 \rightarrow (-4)/(-1) = 4 > 0$. ✓
 - $-2 < x < 1$: $x = 0 \rightarrow (-1)/(2) = -0.5 < 0$. ✗
 - $x > 1$: $x = 2 \rightarrow (1)/(4) = 0.25 > 0$. ✓

$$\rightarrow x < -2 \text{ or } x > 1.$$

Example 6: When the Right Side Isn't 0 — Move Everything and Combine

$$\frac{2}{x-1} \geq 1.$$

- ① Move the 1 on the right to the left $\rightarrow \frac{2}{x-1} - 1 \geq 0$.
- ② Combine into one fraction: $\frac{2-(x-1)}{x-1} = \frac{3-x}{x-1} \geq 0$.
- ③ Numerator 0: $x = 3$. Denominator 0: $x = 1$ (excluded).

- $x < 1$: $(+)/(-) = -$. ✖
- $1 < x \leq 3$: $(+)/(+) = +$. ✓ ($x = 3$ included)
- $x > 3$: $(-)/(+) = -$. ✖

$$\rightarrow 1 < x \leq 3.$$

Example 7: Higher-Degree Inequalities — Tear and Mark Every Root

$$x^3 - 3x^2 - 4x + 12 \geq 0.$$

- ① Try plugging in $x = 2$: $8 - 12 - 8 + 12 = 0$. Exactly 0.
- ② Pull out $(x - 2)$ using synthetic division.
Quotient $x^2 - x - 6 = (x - 3)(x + 2)$.
 $\rightarrow (x - 2)(x - 3)(x + 2) \geq 0$.
- ③ Zero points: $x = -2, 2, 3$. Four intervals.

- $x < -2$: $(-)(-)(-) = -$. ✖
- $-2 < x < 2$: $(-)(-)(+) = +$. ✓
- $2 < x < 3$: $(+)(-)(+) = -$. ✖

- $x > 3$: $(+)(+)(+) = +$. ✓

→ $-2 \leq x \leq 2$ or $x \geq 3$.

Example 8: $(x - a)^n$ — Even Powers Don't Flip the Sign

$$(x - 1)^2(x + 2) < 0.$$

- ① Zero points: $x = 1$ (twice — even), $x = -2$ (once — odd).
- ② Even powers don't change sign when crossing. Only odd powers do.
- ③ Intervals:

- $x < -2$: $(+)^2(-) = -$. ✓
- $-2 < x < 1$: $(+)^2(+) = +$. ✗
- $x > 1$: $(+)^2(+) = +$. ✗

→ $x < -2$.

Example 9: Single Absolute Value — Distance from the Zero Point

$$|x - 3| < 5.$$

- ① An absolute value inequality is "distance left and right from the zero point."
 $-5 < x - 3 < 5$.
- ② Add 3 → $-2 < x < 8$.

$$|2x + 1| \geq 3.$$

- ① Outside the zero point in both directions:
 $2x + 1 \leq -3$ or $2x + 1 \geq 3$.
- ② $2x \leq -4 \rightarrow x \leq -2$. Or $2x \geq 2 \rightarrow x \geq 1$.

$$\rightarrow x \leq -2 \text{ or } x \geq 1.$$

Example 10: Two Absolute Values — Split into Intervals and Handle One by One

$$|x - 1| + |x + 2| \leq 5.$$

- ① Find where each absolute value's inside hits 0: $x = 1$, $x = -2$.
- ② Split the whole line into three using these points.

Interval 1: $x < -2$

Both insides are negative $\rightarrow$ flip signs to strip the absolute bars.

$$-(x - 1) - (x + 2) = -2x - 1 \leq 5.$$

$$-2x \leq 6 \rightarrow x \geq -3.$$

Intersect with the interval $\rightarrow -3 \leq x < -2$.

Interval 2: $-2 \leq x < 1$

$(x - 1)$ negative, $(x + 2)$ positive.

$$-(x - 1) + (x + 2) = 3 \leq 5. \text{ Always true.}$$

$\rightarrow$ The whole interval is an answer: $-2 \leq x < 1$.

Interval 3: $x \geq 1$

Both positive $\rightarrow$ just strip the bars.

$$(x - 1) + (x + 2) = 2x + 1 \leq 5.$$

$$2x \leq 4 \rightarrow x \leq 2.$$

Intersect with the interval $\rightarrow 1 \leq x \leq 2$.

- ③ Combine all three $\rightarrow -3 \leq x \leq 2$.

Part C: Exponential and Logarithmic Inequalities

Example 11: Exponential Inequalities — The Inequality Direction Depends on the Base

$$2^{x+1} > 8.$$

① Write 8 as a power of base 2: $8 = 2^3$.

② $2^{x+1} > 2^3$. Base $2 > 1 \rightarrow$ keep the inequality as is: $x + 1 > 3 \rightarrow x > 2$.

$$\left(\frac{1}{2}\right)^{x+1} \geq \frac{1}{4}.$$

① $\frac{1}{4} = \left(\frac{1}{2}\right)^2$.

② Base $\frac{1}{2} < 1 \rightarrow$ **flip the inequality**: $x + 1 \leq 2 \rightarrow x \leq 1$.

Example 12: Logarithmic Inequalities — Check the Argument First

$$\log_2(x - 1) < 3.$$

① Write 3 as a base-2 log: $3 = \log_2 8$.

② $\log_2(x - 1) < \log_2 8$. Base $2 > 1 \rightarrow x - 1 < 8 \rightarrow x < 9$.

③ **Argument condition**: $x - 1 > 0 \rightarrow x > 1$.

$$\rightarrow 1 < x < 9.$$

$$\log_{\frac{1}{2}}(x + 2) \geq 1.$$

① $1 = \log_{\frac{1}{2}} \frac{1}{2}$.

② Base $\frac{1}{2} < 1 \rightarrow$ flip: $x + 2 \leq \frac{1}{2} \rightarrow x \leq -\frac{3}{2}$.

③ Argument condition: $x + 2 > 0 \rightarrow x > -2$.

$$\rightarrow -2 < x \leq -\frac{3}{2}.$$

Part D: Inequalities Crossing Integer Boundaries

Example 13: First, Get the Meaning of $\lfloor x \rfloor$ Down

$\lfloor x \rfloor$ is the greatest integer that does not exceed x .

Let's pin down some values:

$$\lfloor 3.7 \rfloor = 3, \lfloor 3.0 \rfloor = 3, \lfloor 0.5 \rfloor = 0.$$

$$\lfloor -1.2 \rfloor = -2, \lfloor -1.0 \rfloor = -1, \lfloor 5 \rfloor = 5.$$

Now trace the pattern we just saw. $\lfloor 3.7 \rfloor = 3$ means $3 \leq 3.7 < 4$.

$\lfloor 0.5 \rfloor = 0$ means $0 \leq 0.5 < 1$. $\lfloor -1.2 \rfloor = -2$ means $-2 \leq -1.2 < -1$.

So $\lfloor x \rfloor = 3$ means $\rightarrow 3 \leq x < 4$.

$$\lfloor 2x \rfloor = 5 \rightarrow 5 \leq 2x < 6 \rightarrow 2.5 \leq x < 3.$$

Example 14: Inequalities Containing $\lfloor x \rfloor$ — Replace with t

$$\lfloor x \rfloor^2 - \lfloor x \rfloor - 6 < 0.$$

① Replace $\lfloor x \rfloor$ with t : $t^2 - t - 6 < 0$.

② Tear: $(t - 3)(t + 2) < 0 \rightarrow -2 < t < 3$.

③ $t = \lfloor x \rfloor$ is an integer. $\rightarrow t = -1, 0, 1, 2$.

④ Convert each t value to an x interval:

$$\lfloor x \rfloor = -1 \rightarrow -1 \leq x < 0.$$

$$\lfloor x \rfloor = 0 \rightarrow 0 \leq x < 1.$$

$$\lfloor x \rfloor = 1 \rightarrow 1 \leq x < 2.$$

$$\lfloor x \rfloor = 2 \rightarrow 2 \leq x < 3.$$

$$\rightarrow -1 \leq x < 3.$$

Example 15: $\lceil x + 1 \rceil > \lceil x \rceil$ — Always True?

$$\lceil x + 1 \rceil > \lceil x \rceil.$$

When x is an integer n : $\lceil n + 1 \rceil = n + 1$, $\lceil n \rceil = n \rightarrow n + 1 > n$. True.

When x is not an integer, $x = n + \delta$: $\lceil x + 1 \rceil = n + 1$, $\lceil x \rceil = n \rightarrow n + 1 > n$. True.

→ **True for all x** . Reason: $\lceil x + 1 \rceil = \lceil x \rceil + 1$ always holds.

Example 16: Ceiling Function and Fractional Part

$\lceil x \rceil$: the smallest integer that is not less than x .

$$\lceil 3.2 \rceil = 4, \lceil -1.2 \rceil = -1.$$

$\{x\} = x - \lfloor x \rfloor$: the fractional part of x . Always $0 \leq \{x\} < 1$.

$$\{3.7\} = 0.7, \{-1.2\} = 0.8.$$

Usage:

$$\lceil x \rceil = \lfloor x \rfloor \rightarrow \text{when } x \text{ is an integer.}$$

$$\lceil x \rceil = \lfloor x \rfloor + 1 \rightarrow \text{when } x \text{ is not an integer.}$$

$$\{x\} > 0.5 \rightarrow \text{for every integer } n, \text{ the interval } n + 0.5 < x < n + 1.$$

Common Mistakes

Mistake 1: Recklessly Multiplying Both Sides of a Rational Inequality by the Denominator

Wrong path: " $\frac{x-1}{x+2} > 0$ so multiply by $x + 2$ and get $x - 1 > 0$."

Why it's wrong: You don't know whether $x + 2$ is positive or negative. The inequality direction is

unknown.

Right path: Split into intervals using the points where numerator and denominator each equal 0. Never multiply by the denominator.

Mistake 2: Forgetting the Argument Condition in Logarithmic Inequalities

Wrong path: " $\log_2(x - 1) < 3 \rightarrow x < 9$."

Why it's wrong: A logarithm's argument must be strictly positive. $x - 1 > 0 \rightarrow x > 1$.

Right path: Solve the inequality, then **always intersect with argument > 0** .

Mistake 3: Writing $[x] \geq 2$ as $x \geq 2$ — Coincidentally Right, but Dangerous

Wrong path: " $[x] \geq 2$ so $x \geq 2$." — in this case it happens to be right, but...

Why it's dangerous: $[x] > 2$ actually means $x \geq 3$. Because $[x] = 2$ covers $2 \leq x < 3$. You must check the equals sign at integer boundaries every time.

Right path: Always start from $[x] = n \leftrightarrow n \leq x < n + 1$.

Mistake 4: Thinking the Sign Flips When Crossing an Even-Power Factor

Wrong path: " $\ln(x - 1)^2(x + 2)$, the sign flips at $x = 1$ too."

Why it's wrong: $(x - 1)^2$ is 0 at $x = 1$, but because it's squared, its sign is always positive. On both sides of $x = 1$, $(x - 1)^2 > 0$. No sign change.

Right path: $(x - a)^{\text{even}} \rightarrow$ sign unchanged. Only $(x - a)^{\text{odd}}$ flips the sign.

What We Just Did

- ① Make the right side 0. Tear the left side completely.
- ② Mark every zero point on the number line. Always exclude denominator=0.
- ③ Pick one number in each interval and judge the sign of the torn expression.
- ④ Remember: even-power factors don't flip the sign.
- ⑤ Exponentials/logs: base>1 → keep inequality, 0<base<1 → flip.
For logs, always intersect with argument>0.
- ⑥ Absolute values: split into intervals at the zero points, strip bars one by one.
- ⑦ [x]: substitute t=[x] first. Remember t is an integer.

Exercise 1

$\frac{x^2-4}{x+3} \leq 0$. Tear the numerator, mark denominator $\neq 0$, split into intervals, judge signs.

→ Follow: **Example 5**

Solutions: [Solution Set](#)

Exercise 2

$2^x + 2^{x+1} > 48$. Start by pulling out 2^x .

→ Follow: **Example 11**

Solutions: [Solution Set](#)

Exercise 3

$\log_2(x^2 - 3x) \leq 2$. Make sure to intersect with the argument > 0 condition.

→ Follow: **Example 12**

Solutions: [Solution Set](#)

Exercise 4: Constructive

Find the range of x that satisfies $[x]^2 - 5[x] + 6 = 0$.

Then make your own quadratic equation in $[x]$ of the same form and solve it.

→ Follow: **Example 14**

Solutions: [Solution Set](#)

Exercise 5

$|x - 2| + |x + 3| \leq 7$. Split into three intervals at the zero points.

→ Follow: **Example 10**

Solutions: [Solution Set](#)

Exercise 6: Challenge

Solve $\{x\} + [x]^2 \leq x$ for $0 \leq x < 3$.

Substituting $\{x\} = x - [x]$ gives a simple form.

→ Follow: **Examples 14, 16**

Today's Procedure

Step 1: Make the right side 0. Tear the left side completely.

Step 2: Mark every zero point on the number line. Exclude denominator=0.

(Watch for even-power factors – sign doesn't flip there.)

Step 3: Pick one number per interval and check the sign.

- Rational: check numerator and denominator signs separately
- Exponential/log: base>1 keep, 0<base<1 flip. Logs: intersect with argument>0
- n absolute values: split at zero points, handle one interval at a time
- [x]: substitute $t=[x]$; t is an integer

Terminology

Up to now, we've only used simple words: "tear", "move", "mark on the number line", "check the sign", "split".

You already know the methods. Now we give them their math names.

What we've been calling it	Math Term	Symbol / Explanation
the inequality sign flips	inequality direction reversal	when multiplying by a negative or base<1
point where it hits 0	critical point	critical point
splitting into intervals	interval method	interval method
checking only the sign of the torn expression	sign chart	sign chart
discriminant	discriminant	$D = b^2 - 4ac$
argument	argument	the expression inside a

What we've been calling it	Math Term	Symbol / Explanation
		logarithm
base	base	base
floor function	floor function (Gauss notation)	$\lfloor x \rfloor = [x]$
ceiling function	ceiling function	$\lceil x \rceil$
fractional part	fractional part	$\{x\} = x - [x]$
absolute value	absolute value	$ x $